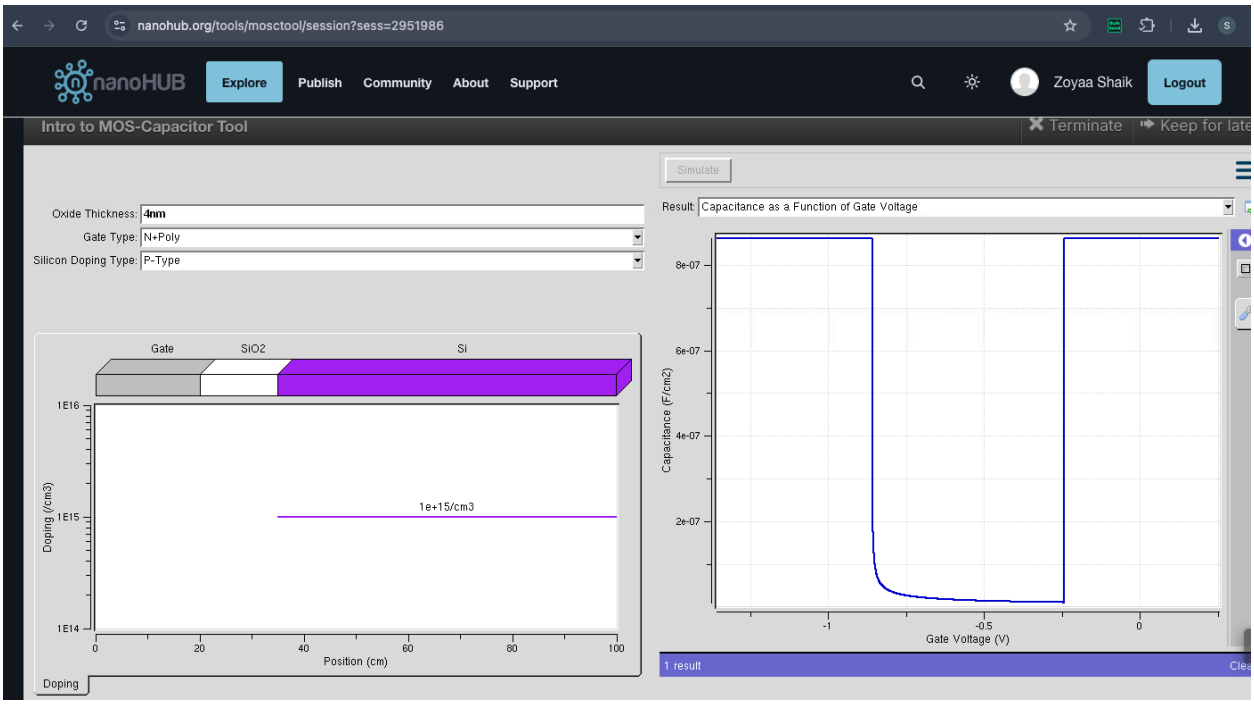
EXPERIMENT 3

Oxide Thickness: 4nm

Gate Type: N+Poly

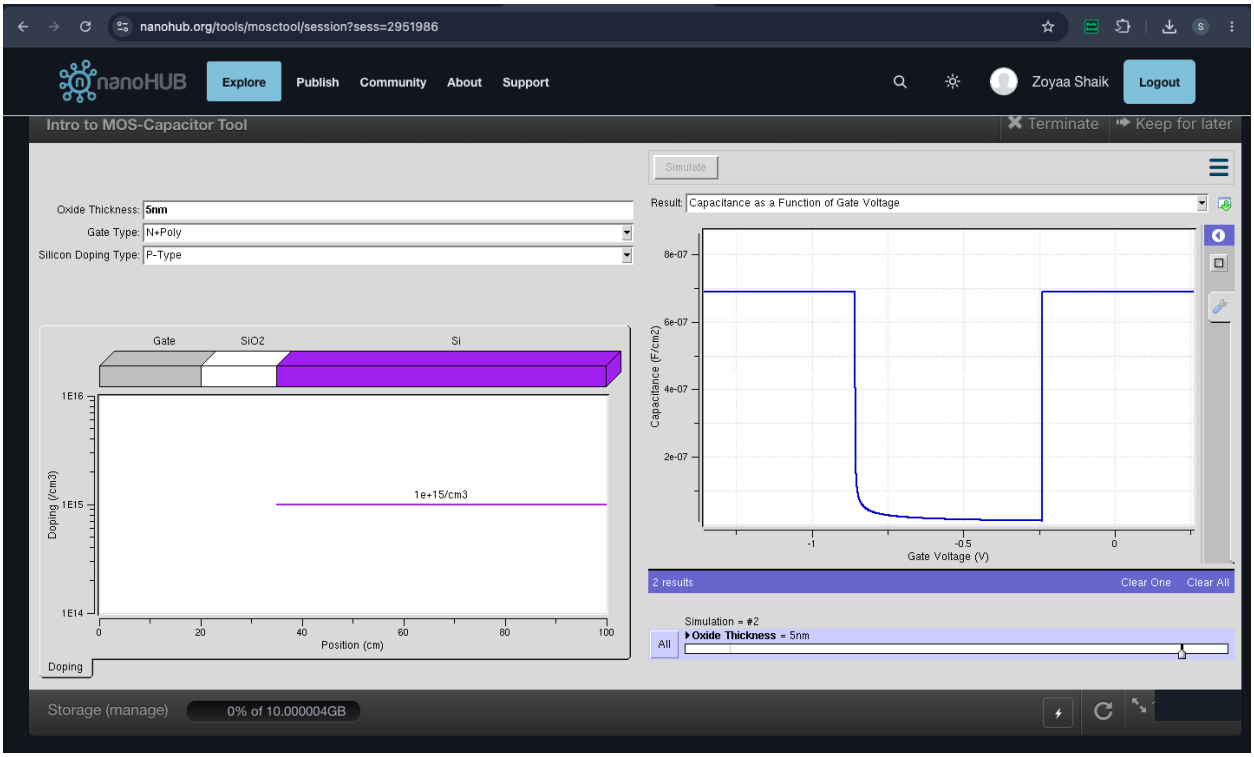
Silicon Doping Type: P-Type Poly



Oxide Thickness: 5nm

Gate Type: N+Poly

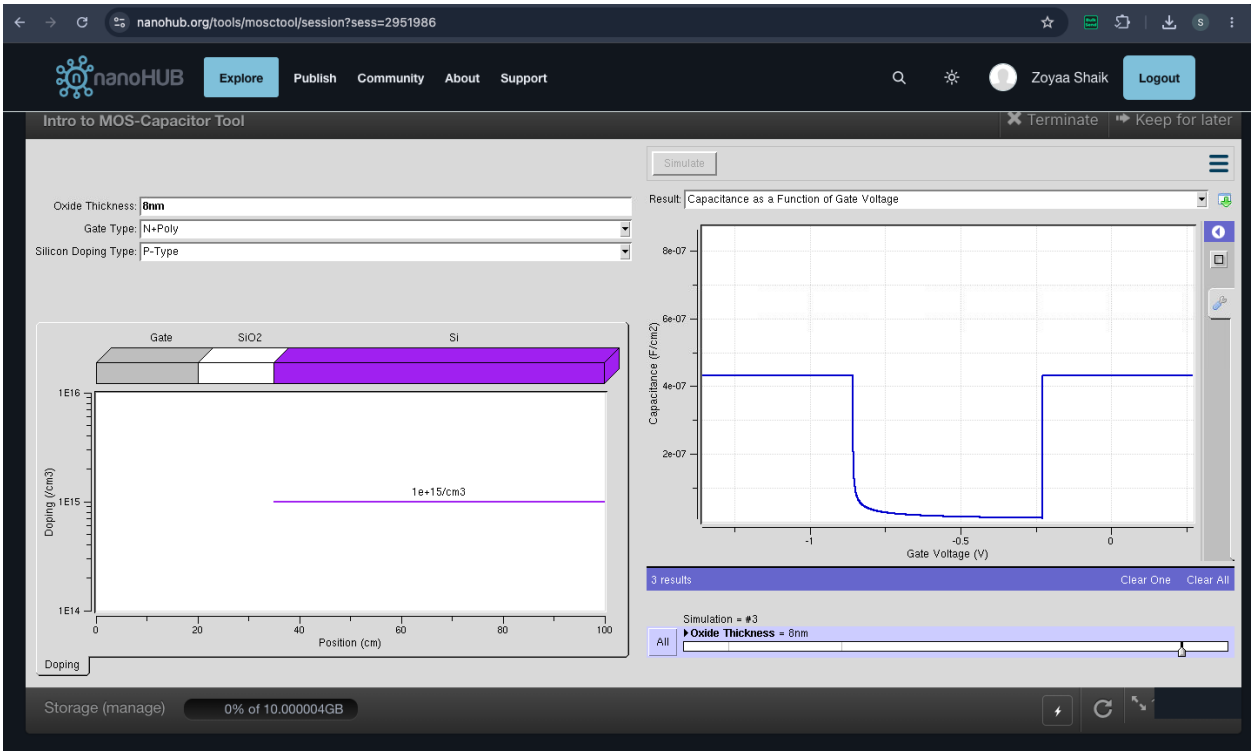
Silicon Doping Type: P-Type Poly



Oxide Thickness: 8nm

Gate Type: N+Poly

Silicon Doping Type: P-Type Poly



Effect of Oxide Thickness (t_ox) on MOS Capacitance

* 4 nm Oxide Thickness:
  - Maximum Accumulation Capacitance (C_ox): ~8.6 x 10^-7 F/cm^2
  - Highest capacitance among the three runs.

* 5 nm Oxide Thickness:
  - Maximum Accumulation Capacitance (C_ox): ~6.9 x 10^-7 F/cm^2
  - Capacitance drops by approximately 20% relative to 4 nm.

* 8 nm Oxide Thickness:
  - Maximum Accumulation Capacitance (C_ox): ~4.3 x 10^-7 F/cm^2
  - Capacitance is reduced to half of the value obtained at 4 nm.

Explanation:

The oxide layer acts as a parallel-plate dielectric between the gate electrode and the silicon substrate. The oxide capacitance per unit area is inversely proportional to its thickness:

C_ox = epsilon_ox / t_ox

As oxide thickness increases from 4 nm to 8 nm, the distance between the gate and substrate increases, reducing the charge storage capacity for a given voltage. Consequently, maximum accumulation capacitance decreases inversely with t_ox.